IPUMS DHS

VEGETATIVE INDEX AND LAND COVER DATA OVERVIEW

An almost overwhelming number of vegetative and land cover data types exist. This post features a few common indices and land cover data types, and gives a brief primer into which ones might be most appropriate for different research goals.

NDVI LAI VCF WRSI LAND COVER AGRICULTURE SEASONALITY TERRA SF

AUTHOR
Devon Kristiansen

AFFILIATION
IPUMS Global Health Research Manager

PUBLISHED
January 7, 2026

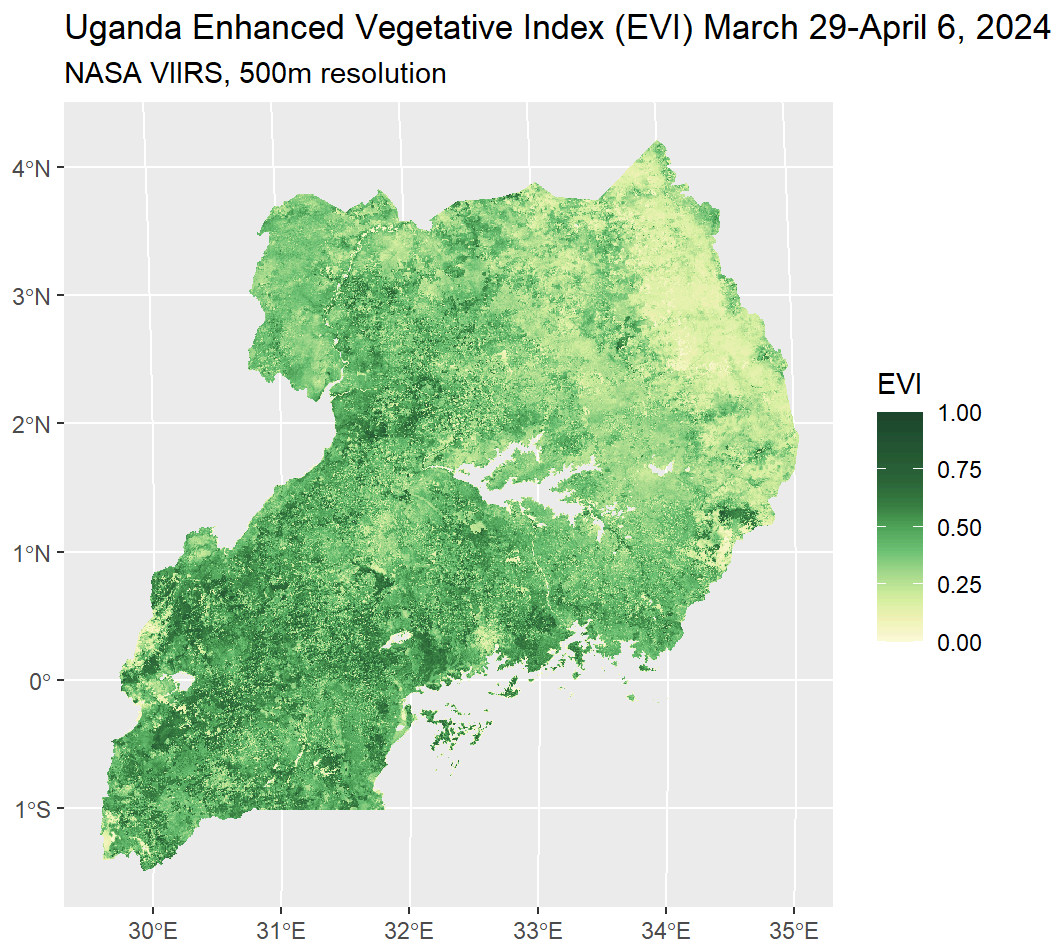
Several previous posts on this blog (iterative workflows; NDVI aggregation methods; multilevel modeling) have featured the Normalized Difference Vegetation Index (NDVI), which is a measure of vegetation greenness and health.

There are many other measures to describe vegetation or land cover on Earth's surface, however, and in this post we'll go into details about what these data are and how they might be appropriate for certain research goals. The various indicators could represent seasonal agricultural productivity, or be used to measure such droughts, deforestation, or many other climate change-related phenomena. We hope this blog helps you get started with your research project!

GREENNESS OR VEGETATION

First, we'll look more closely at measures related to the NDVI, in that the primary method to produce the measure is a calculation of the density of greenness, or photosynthetic activity, of a pixel. Below is an image we've created of the Enhanced Vegetation Index of Uganda in 2024 using remotely-sensed data from NASA. Click the word "Code" below to see the entirety of the code used to produce this image.

▶ Code



The table below is not an exhaustive list. We've chosen a select set of indicators to highlight that are widely used and vary enough from each other to be considered for different research goals. A longer list of vegetation indices can be found on NASA's website.

The information in the table is designed to help you navigate which indicator might be most appropriate for your research question. Your choice might be guided by the spatial resolution, temporal frequency, spatial or temporal coverage, or specific component of the measure itself. You will notice some indicators occupy multiple rows - in the cases that the same indicator was collected by different instruments (ie. satellites) over time, their resolution or frequency might have changed as technology in remote sensing has advanced. Below the table are additional details regarding each measure and suggestions or examples for best use cases.

Index or measure	Instrument	Basic content	Temporal extent	Frequency	Resolution	Where to download
NDVI	AVHRR	Greenness and vegetation health	1981 - present	16-day	0.05 degree (~5km at equator)	NCEI NOAA direct download

Index or measure	Instrument	Basic content	Temporal extent	Frequency	Resolution	Where to download
NDVI	MODIS	Greenness and vegetation health	2000-2025	16-day	250m	NASA Earthdata Search - MOD13Q1
NDVI	VIIRS	Greenness and vegetation health	2012 - present	16-day	500m	NASA Earthdata Search - VNP13A1
EVI	VIIRS	Greenness and vegetation health	2012- present	16-day	500m	NASA Earthdata Search - VNP13A1
EVI	Landsat	Greenness and vegetation health	December 2024 - present	2-3 days	30m	NASA Earthdata Search - HLS Operational Land Imager Vegetation Indices Daily Global 30 m V2.0
VCF	AVHRR	Percent of cell covered in tree, non-tree vegetation, and non-vegetation	1982-2016	annual	0.05 degree (~5km at equator)	NASA Earthdata Search - VCF5KYR
VCF	MODIS	Percent of cell covered in tree, non-tree vegetation, and non-vegetation	2000-2021	annual	250m	NASA Earthdata Search - MOD44B
LAI	AVHRR	Canopy/leaf cover	1981 to 10 days before the present	daily	0.05 degree (~5km at equator)	NCEI NOAA direct download
LAI	MODIS	Canopy/leaf cover	2002 - present (2000 - present)	4-day (8-day)	500m	NASA Earthdata Search - MCD15A3H (MOD16A2GF)
LAI	VIIRS	Canopy/leaf cover	2012 - present	8-day	500m	NASA Earthdata Search - VNP15A2H

NDVI

Measurement

As we have covered previously in our introductory post on NDVI, the Normalized Difference Vegetation Index represents the amount of photosynthetic activity measured by a satellite on a pixel of Earth's surface. Because photosynthetic plants absorb red light and reflect greens, the difference between the near infrared (NIR) and red (R) light reflected back to satellite sensors, normalized by the sum of the near infrared and red bands of light.

$$NDVI = (NIR - R)/(NIR + R)$$

The possible values of NDVI range from -1 to 1, with values zero and under being bare ground, water, and other surfaces that reflect large amounts of red light relative to near infrared light, that is, having very little to no plants absorbing the red light for photosynthesis.

Typical Uses

The NDVI is a widely-used measurement for vegetation health, and can be used to detect changes in vegetation cover, density, and stress over time.

Some noted weaknesses of NDVI are that it can be prone to containing outliers and spurious value changes in the data due to cloud cover, and the reflectance of different soil types may interfere with its ability to compare pixels from different regions. NDVI has been found to not perform well differentiating pixels that have a high vegetation density, such as tropical rain forest. Further, not all photosynthetic surfaces of plants are green[1,2].

One option for research projects involving areas with low vegetative cover is the Soil Adjusted Vegetation Index (SAVI), which is the NDVI corrected for soil brightness.

Example articles

- Lopez-Carr et al 2016 used NDVI and precipitation data to analyze child nutrition in the Lake Victoria basin[3].

EVI

Measurement

The Enhanced Vegetation Index is similar to the NDVI, but with additional components to adjust for cloud cover and canopy background noise.

$$EVI = G * ((NIR - R)/(NIR + C1 * R – C2 * B + L))$$

EVI is calculated from the above equation from near infrared light (NIR), red light (R), blue light (B), a

canopy background value for the pixel (L), coefficients for atmospheric resistance (C1, C2), and a gain factor (G - usually 2.5).

Typical Uses

The EVI might be a better choice over NDVI for an analysis focusing on areas of the world with frequent cloud cover and dense foliage (e.g. tropical rain forest) or smoke in the atmosphere (e.g. slash and burn agriculture). EVI is available through NASA starting in 2012, so it is not a viable measurement option for research questions involving a timeframe before 2012.

Example article

- Davies et al 2016 found that EVI was more resilient to atmospheric interference than other vegetative indices when measuring forest changes in Cambodia[4].

LAI

Measurement

The Leaf Area Index is a measurement of the green leaf density of plant canopy, or the one-sided leaf area per horizontal ground area. There are multiple approaches, but the end result is a unit-less index value between zero and 1[5,6].

- Leaf Area (m^2) / Ground Area (m^2)
- Integral of leaf area over the canopy height (m^3/m^3)
- One-sided green broadleaf area per unit of ground area (m^2/m^2); one-half of the total conifer needle surface area per unit of ground area

Typical Uses

The LAI is a reasonable choice to represent a variety of biological processes and activity of plant matter in an area, such as photosynthetic production, rainfall interception, or evapotranspiration[7].

Example article

- Bangelesa et al 2023 found an association between LAI and stunting in children under 5 in the Democratic Republic of the Congo[7].

VCF

Measurement

The Vegetation Continuous Fields data are produced by NASA to describe the composition of land cover within each pixel, divided into tree cover, non-tree vegetative cover, and non-vegetative cover, such that

the sum of the three components is 100 percent.

The data are produced, like the other vegetative indices, using surface reflectance (or light captured by satellite reflected off Earth's surface), but combined with training data to create the model for predicting global land coverage in the three categories.

Typical Uses

The VCF data may be useful for research projects if the outcome may be correlated differently between tree and non-tree vegetative cover, which is difficult to determine using NDVI. VCF may be more useful than land cover data if the density of forest cover relative to non-tree vegetative cover is meaningful.

Because the VCF data are an annual-based measure, VCF might not be well suited to study change within the year, or specific deforestation events with a narrow time scale. For example, VCF would not be suited to measuring the loss of vegetative cover immediately after a fire, as some amount of vegetation would grow back over a period of months until the next measurement occurs.

Example article

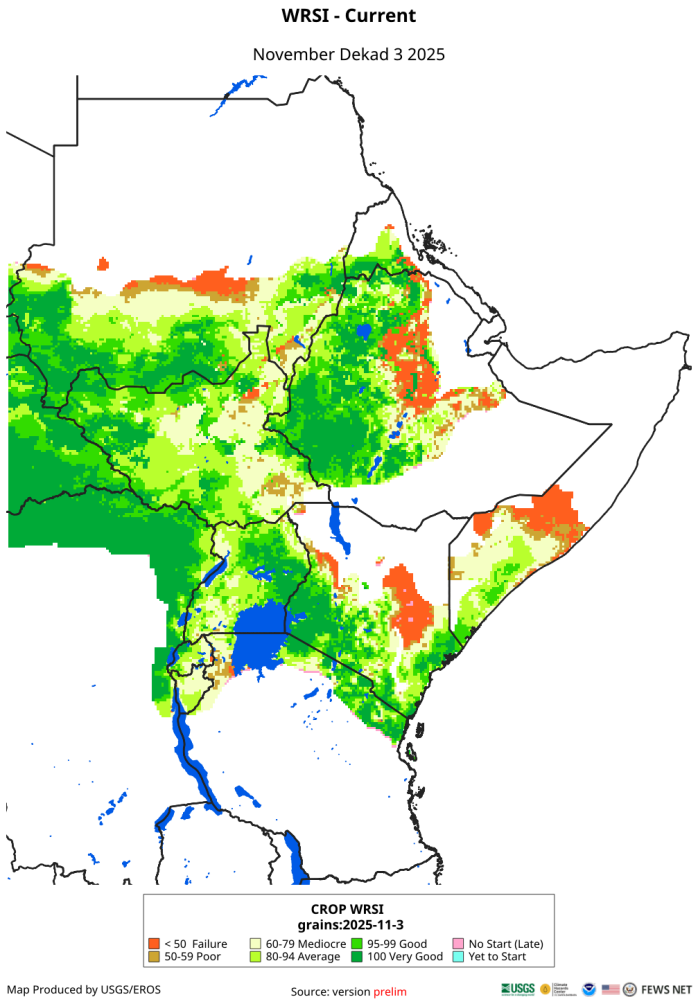
- Johnson et al 2013 used VCF data to demonstrate a linkage between percent forest cover and reduced risk of diarrhea in young children, and that children living in areas with deforestation in the past 10 years were less likely to consume foods rich in vitamin A or meet dietary diversity guidelines[8].

HONORABLE MENTIONS

- The Normalized Difference Fraction Index has been demonstrated as a very accurate measure of deforestation and forest degradation, but it is not available at a global scale for download. Instead, one would need to download the component pieces of surface reflectance data and compute NDFI for the area of focus[2,9].

ENVIRONMENTAL MOISTURE MEASURES

Beyond greenness of foliage, remotely sensed data can be used to indicate vegetation stress as it relates to precipitation. Below are a few key indicators you might come across. Below is a map of east Africa indicating the Water Satisfaction Requirement Index for the last ten days of November 2025 from the FEWSNET program.



Current WRSI of East Africa from USGS FEWSNET, November 21-30, 2025

Index or measure	Instrument	Basic content	Temporal extent	Frequency	Resolution	Where to download
WRSI	MERRA2 (reanalysis of satellite data from NASA); CHIRPS	Extent to which rainfall meets crop water demands	2001 - present	10-day	0.1 degree (~10km at the equator)	FEWSNET
SWI	FAO Digital Soil Map of the World, CHIRPS, and MERRA2	Water retained in soil as percent of full water holding capacity	2001 - present	10-day	0.1 degree (~10km at the equator)	FEWSNET
NDMI	Landsat (HLS)	Vegetation water content	2013 - present	daily	30m	On-demand order ESPA USGS

WRSI

Measurement

The Water Requirement Satisfaction Index (WRSI) uses historic information of crops typically grown in a region and the soil type combined with remotely sensed precipitation and evapotranspiration data to approximate crop performance and/or water stress. The index generally ranges from 0 to 100, but can be greater than 100 when rainfall exceeds the crop water requirements. A value under 50 indicates significant drought or crop failure.

Note that the index is relative to a specific crop's water demand, and WRSI data can currently be downloaded only for regions in Africa as part of the Famine Early Warning System project from the US Geological Survey (USGS).

Typical Uses

The WRSI is useful for analyses aiming to measure crop yields and agricultural production. This is particularly helpful in that any given region's crop harvest quality may differ from another region even with the same amount of precipitation if the soil holds less moisture or if one region's primary crop is more water-sensitive. The 10-day frequency of WRSI is a fine enough temporal resolution to assist in precise drought detection.

WRSI is usually calculated for one or two common crops for certain regions, so it is a representation of harvests at a regional level, but not necessarily pixel by pixel, for example, in the case that cultivators rotate crops or choose not to cultivate in a field for a particular season.

Example article

- Grace et al 2023 used the WRSI to measure seasonality of growing seasons, and found an association between growing season quality and reproductive health decision-making[10].

SWI

Measurement

The Soil Water Index is a measure of the water retained in soil based on a number of inputs. The final calculation is the amount of water in the soil (SW) divided by the water holding capacity (WHC), multiplied by 100. The possible values are between 0 and 100.

$$SWI = (SW/WHC) * 100$$

Soil water content at time i (SW_i) is measured by the known soil water content in the previous time frame, plus precipitation in time i (PPT_i) minus the potential evapotranspiration during time i (PET_i).

$$SW_i = SW_{i-1} + PPT_i - PET_i$$

The WHC is determined using known qualities about the water retention qualities of different soil types, and the spatial extent of soil types comes from FAO's Digital Soil Map of the World.

Typical Uses

The SWI could be a valuable indicator to measure disease vectors such as mosquitos spreading malaria, detecting floods in addition to drought, or to assist in measuring the water stress on vegetation that is not a specific crop, such as rangelands for pastoral uses.

Example article

- Zbiri et al 2019 categorized areas of Morocco according to NDVI and calculate drought indicators using the SWI[11].

NDMI

Measurement

Like NDVI, NDMI uses surface reflectance to measure vegetation quality using a normalized ratio of light bands. Instead of red light, NDMI uses shortwave infrared (SWIR) and near infrared (NIR).

$$NDMI = (NIR - SWIR)/(NIR + SWIR)$$

Detecting the difference between NIR and SWIR indicates the water content within plant matter. The index ranges from -1 to 1, with zero indicating moderate moisture, values below zero indicating water stress or sparse vegetation, and values above zero indicating more vegetative cover and higher water content.

Typical Uses

NDMI can be used for indicators of drought, wildfire risk, and to inform irrigation management. The index measures leaf moisture regardless of whether the vegetation is a tree or not. The weakness of NDMI is that it is difficult to distinguish low water content from sparse vegetative cover and is often used with other remotely sensed data such as NDVI to indicate greenness or density of vegetative cover.

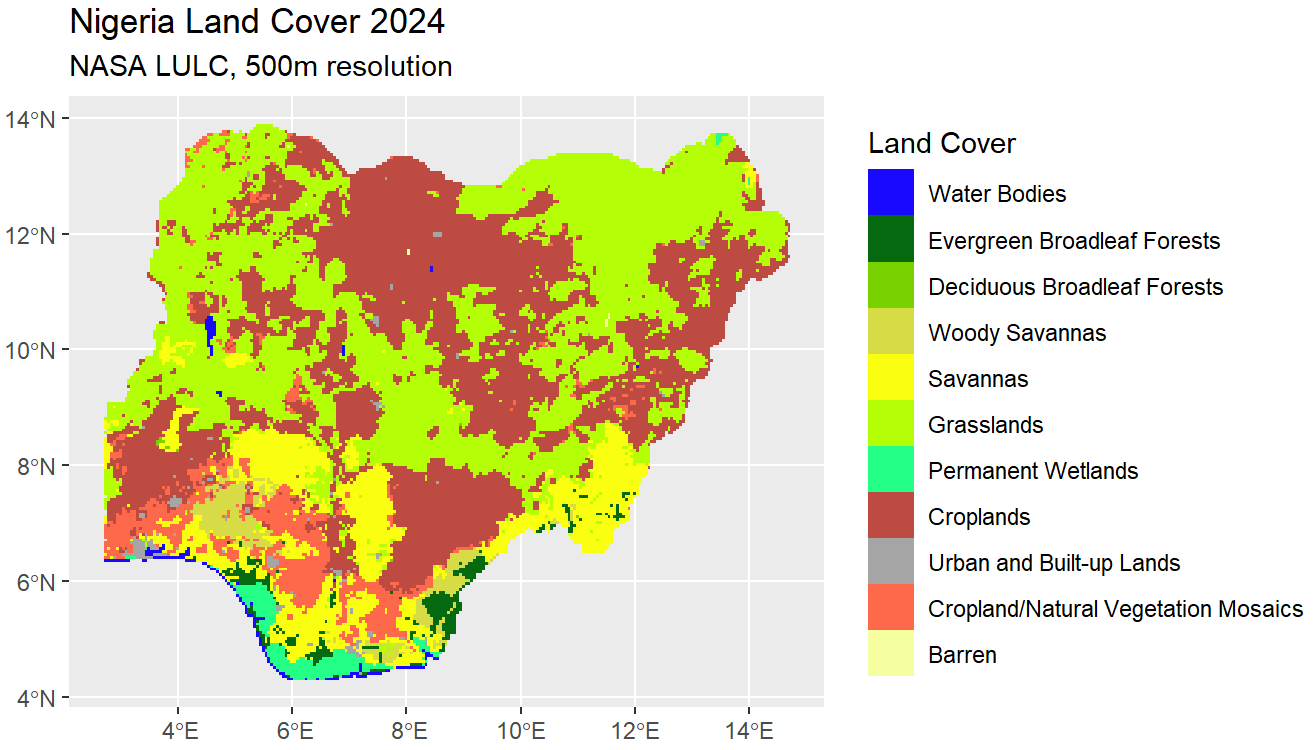
Example article

- Moisa et al 2025 leveraged NDMI and NDVI data together with Land Use/Land Cover (described below) to assess climate change resilience outlooks in Ethiopia, noting that land converted from forest to cropland contained less moisture, suggesting future water resource challenges[12].

LAND COVER

Obviously, vegetative cover is not the only type of surface on Earth. Datasets such as Land Use/Land Cover from NASA and Global Dynamic Land Cover from the Copernicus Programme describe the human lived environment using classification schemes that describe environments in terms of impervious surfaces (urban or built up) or by ecosystem type (wetland, evergreen forest, savannah, permanent ice cover, etc). Below is a map using the LULC data for Nigeria in 2024. Click "Code" below to see the R code that created this map.

▶ Code



Index or measure	Instrument	Basic content	Temporal extent	Frequency	Resolution	Where to download
LULC	MODIS (Terra+Aqua)	Land cover classification schemes up to 17 categories	2001 - 2021	annual	500m	NASA Earthdata Search - MCD12Q1/MCD12C1 (new version)

Index or measure	Instrument	Basic content	Temporal extent	Frequency	Resolution	Where to download
LULC	Landsat (GFSAD)	Land cover classification schemes - various number of classes	2007 - 2016 (varies by region)	multi-year	30m - 1km	NASA Earthdata Search - GFSAD30
LULC	Landsat (GLanCE)	Land cover classification scheme with 7 categories	2001 - 2019	annual	30m	NASA Earthdata Search - Global Land Cover Mapping and Estimation Yearly 30 m V001
GDLC	Copernicus Sentinel-1 and Sentinel-2	11 classes of land cover with multiple forest classes	2015 - 2020	annual	100m (10m for 2020)	Copernicus Programme

LULC

Measurement

NASA's Land Use/Land Cover data represent pixels of Earth's surface using several land use classification schemes. Land cover/land use data describe human's lived environment, in both biophysical (such as woody savannah) and socioeconomic features (urban or built up land).

LULC data is available in standardized schemes with global coverage, but also at finer spatial resolution with more specific land cover types for smaller regions. Surface reflectance is the key component of how land cover at a pixel level is determined, but other inputs are used, especially in regional data.

Typical Uses

Land Use/Land Cover data have a wide variety of applications, such as studying climate change, land use change dynamics, deforestation, urbanization, and more. The categories distinguish between different types of ecosystems, which is not easily accomplished with some of the vegetative indices described earlier. These data distinguish human-built impervious surfaces from bare ground/no vegetative cover.

The wide scope of spatial resolution, temporal extent, and varying number of categories make the different LULC data flexible for many applications. The exceptions being change within a year or measurements regarding health of vegetation, which are better suited for the vegetative indices described above.

Example articles

- Ganem et al 2025 uses regional LULC data to analyze land cover change in Brazil[13].
- Zaldo-Aubanell et al 2021 is a review article comparing different LULC approaches[14].

GDLC

Measurement

Global Dynamic Land Cover (GDLC) is land use/land cover data from the Copernicus Program, out of the European Space Agency, using Sentinel satellites. The GDLC contain 11 different land cover classes, and create layers with a specific focus on tropical forest types at high spatial resolution. Data inputs include the Digital Elevation Model, land training data, and agro-meteorological indicators.

Typical Uses

Similar to LULC, GDLC can also be used to measure land cover change dynamics, deforestation, climate change, and urbanization. GDLC is less frequent than LULC and covers a narrower time extent, but often has a finer spatial resolution. You may consider using GDLC data for focused regional projects because of the small spatial resolutions, and the distinctions drawn between tropical vegetation types in their pantropical tree cover density data. The small time frame limits its use for understanding change over time, however.

HONORABLE MENTIONS

- Google Earth Engine
- Livelihood Zones from FEWSNET - See Grace and Nagle 2015[15]

References

1. Huete, A., Didan, K., Miura, T., Rodriguez, E. P., Gao, X., & Ferreira, L. G. (2002). Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment*, *83*(1), 195–213. https://doi.org/10.1016/S0034-4257(02)00096-2
2. Schultz, M., Clevers, J. G. P. W., Carter, S., Verbesselt, J., Avitabile, V., Quang, H. V., & Herold, M. (2016). Performance of vegetation indices from Landsat time series in deforestation monitoring. *International Journal of Applied Earth Observation and Geoinformation*, *52*, 318–327. https://doi.org/10.1016/j.jag.2016.06.020
3. López-Carr, D., Mwenda, K. M., Pricope, N. G., Kyriakidis, P. C., Jankowska, M. M., Weeks, J., Funk, C., Husak, G., & Michaelsen, J. (2016). Climate-Related Child Undernutrition in the Lake Victoria Basin: An Integrated Spatial Analysis of Health Surveys, NDVI, and Precipitation Data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *9*(6), 2830–2835. https://doi.org/10.1109/JSTARS.2016.2569411
4. Davies, K. P., Murphy, R. J., & Bruce, E. (2016). Detecting historical changes to vegetation in a Cambodian protected area using the Landsat TM and ETM+ sensors. *Remote Sensing of Environment*, *187*, 332–344.

https://doi.org/10.1016/j.rse.2016.10.027

5. Weiss, M., Baret, F., Smith, G. J., Jonckheere, I., & Coppin, P. (2004). Review of methods for in situ leaf area index (LAI) determination: Part II. Estimation of LAI, errors and sampling. *Agricultural and Forest Meteorology*, *121*(1), 37–53. https://doi.org/10.1016/j.agrformet.2003.08.001
6. Fang, H., Baret, F., Plummer, S., & Schaepman-Strub, G. (2019). An Overview of Global Leaf Area Index (LAI): Methods, Products, Validation, and Applications. *Reviews of Geophysics*, *57*(3), 739–799. https://doi.org/10.1029/2018RG000608
7. Bangelesa, F., Hatløy, A., Mbunga, B. K., Mutombo, P. B., Matina, M. K., Akilimali, P. Z., Paeth, H., & Mapatano, M. A. (2023). Is stunting in children under five associated with the state of vegetation in the Democratic Republic of the Congo? Secondary analysis of Demographic Health Survey data and the satellite-derived leaf area index. *Heliyon*, *9*(2). https://doi.org/10.1016/j.heliyon.2023.e13453
8. Johnson, K. B., Jacob, A., & Brown, M. E. (2013). Forest cover associated with improved child health and nutrition: Evidence from the Malawi Demographic and Health Survey and satellite data. *Global Health: Science and Practice*, *1*(2), 237–248. https://doi.org/10.9745/GHSP-D-13-00055
9. Souza, C., Tenneson, K., Dilger, J., Wespestad, C., & Bullock, E. (2024). Forest Degradation and Deforestation. In J. A. Cardille, M. A. Crowley, D. Saah, & N. E. Clinton (Eds.), *Cloud-Based Remote Sensing with Google Earth Engine: Fundamentals and Applications* (pp. 1061–1091). Springer International Publishing. https://doi.org/10.1007/978-3-031-26588-4_49
10. Grace, K., Kristiansen, D., Boyle, E. H., & Luetke, M. (2023). Investigating Seasonal Agriculture, Contraceptive Use, and Pregnancy in Burkina Faso. *The Professional Geographer*. https://www.tandfonline.com/doi/full/10.1080/00330124.2023.2199316
11. Zbiri, A. (2019). *Drought monitoring using soil water index and normalized difference vegetation index time series in Moroccan rangelands*.
12. Moisa, M., Roba, Z., Purohit, S., Deribew, K., & Gemeda, D. (2025). Evaluating the impact of land use and land cover change on soil moisture variability using GIS and remote sensing technology in southwestern Ethiopia. *Environmental Monitoring and Assessment*, *197*. https://doi.org/10.1007/s10661-025-14301-1
13. Ganem, K. A., Xue, Y., Dutra, A. C., Pareyn, F. G. C., & Shimabukuro, Y. E. (2025). From rainforests to drylands: A context-specific framework for mapping land use and land cover dynamics in Northeast Brazil (2000–2020). *GIScience & Remote Sensing*, *62*(1), 2510140. https://doi.org/10.1080/15481603.2025.2510140
14. Zaldo-Aubanell, Q., Serra, I., Sardanyés, J., Alsedà, L., & Maneja, R. (2021). Reviewing the reliability of Land Use and Land Cover data in studies relating human health to the environment. *Environmental Research*, *194*, 110578. https://doi.org/10.1016/j.envres.2020.110578
15. Grace, K., & Nagle, N. N. (2015). *Using High-Resolution Remotely Sensed Data to Examine the Relationship Between Agriculture and Fertility in Mali*. https://www.tandfonline.com/doi/pdf/10.1080/00330124.2015.1032899?casa_token=jSe8BslY86kAAAAA:RyYtxFCNK0FJ3sxROURFl6-Ax6utI4G8R4AqyLa3J52v94wAulwcrP_17Cs7vdr4mBG3rH25PmSNQg

Developed by IPUMS at the University of Minnesota
Licensed under the Mozilla Public License Version 2.0